the example of H_2O. It can be calculated using the formula

Mass % of an element

$$= \frac{\textit{mass of that element in the compound}}{\textit{molecular mass of the compound}} \times 100$$

Now,

molecular mass of $H_2O = 2(1) + 16$

$= 18$ g

$$\text{Mass \% of hydrogen} = \frac{2}{18} \times 100$$

$= 11.11$ %

$$\text{Mass \% of oxygen} = \frac{16}{18} \times 100$$

$= 88.89$ %

This percentage composition is useful to determine the empirical formula and molecular formula.

Example 1: Find the mass percentage composition of methane (CH_4).

molecular mass of $CH_4 = 12 + 4$

$= 16$ g

$$\text{Mass \% of carbon} = \frac{12}{16} \times 100$$

$= 75$ %

$$\text{Mass \% of hydrogen} = \frac{4}{16} \times 100$$

$= 25$ %

7.6 AVOGADRO HYPOTHESIS

In 1811 Avogadro framed a hypothesis based on the relationship between the number of molecules present in equal volumes of gases in different conditions.

The Avogadro's law states that ***"equal volumes of all gases under similar conditions of temperature and pressure contain equal number of molecules"***

It follows that the volume of any given gas must be proportional to the number of molecules in it. If 'V' is the volume and 'n' is the number of molecules of a gas, then Avogadro law is represented, mathematically, as follows:

$V \alpha n$

$V = \text{constant} \times n$

Thus, one litre ($1\ dm^3$) of hydrogen contains the same number of molecules as in one litre of oxygen, i.e. the volume of the gas is directly proportional to the number of molecules of the gas.

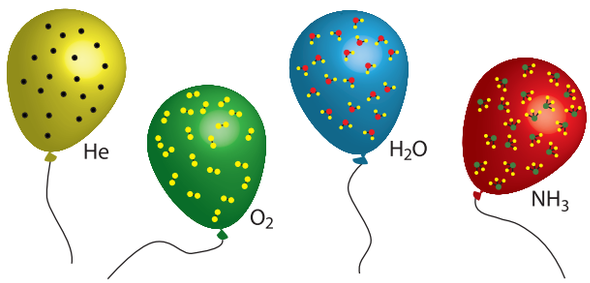


Figure 7.4 Avogadro Hypothesis

Explanation

Let us consider the reaction between hydrogen and chlorine to form hydrogen chloride gas

$H_{2(g)} + Cl_{2(g)} \rightarrow 2\ HCl_{(g)}$

1 vol + 1 vol → 2 volumes

According to Avogadro's law 1 volume of any gas is occupied by "n" number of molecules.

n molecules + n molecules → 2n molecules

if n = 1 then

1molecule + 1 molecule → 2 molecules.

½ molecule + ½ molecule → 1 molecule

1 molecule of hydrogen chloride gas is made up of ½ molecule of hydrogen and ½ molecule of chlorine. Hence, the molecules can be subdivided. This law is in agreement with Dalton's atomic theory.

Activity 7.3

Under same conditions of temperature and pressure if you collect 3 litre of O_2, 5 litre of Cl_2 and 6 litre of H_2,

i. Which has the highest number of molecules?

ii. Which has the lowest number of molecules?

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